

Real-Time Systems / Real-Time Operating Systems

ECE445M/ECE380L.12, Spring 2024

Final Exam

Date: May 4, 2024

UT EID: _____

Printed Name: _____
Last, First

Your signature is your promise that you have not cheated and will not cheat on this exam, nor will you help others to cheat on this exam:

Signature: _____

Instructions:

- Open book, open notes and open web.
- No electronic devices other than your laptop/PC (cell phones off and stowed away).
- You are allowed to access any resource on the internet, but no electronic communication other than with instructors.
- Please be sure that your answers to all questions (and all supporting work that is required) are contained in the space (boxes) provided.
- For all questions, unless otherwise stated, find the most efficient (time, resources) solution.

Problem 1	20	
Problem 2	20	
Problem 3	20	
Problem 4	20	
Problem 5	20+10	
Total	100+10	

Name: _____

Problem 1 (20 points): Miscellaneous

a) Why can an SVC call not use *OS_Wait()*?

b) Should a remote interpreter use a TCP or UDP network connection and why?

c) Explain the difference between external and internal fragmentation? How can you reduce the amount of external and internal fragmentation in a file system?

d) What additional hardware is required to implement virtual memory? Why is this hardware needed for virtual memory?

Name: _____

Problem 2 (20 points): CAN Networking

Given a CAN bus with 3 devices each trying to send a message (with a standard 11 bit ID):

Device	ID
Device 1	0b00110101100
Device 2	0b00110010011
Device 3	0b01000000000

- a) Fill out the CAN trace for the first frame that would end up on the bus and label all the bits. Some of the sections and labels are filled in for you. Hint: note that CAN applies bit stuffing for all bits in the frame, where one stuffed bit is pre-marked below.

Label	SOF												RTR	IDE			
Bit #	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Value	0												0	0	0	0	0

Label	Stuff											CRC					
Bit #	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Value	1	1	0	1	1	0	0	0	0	0	1	1	1	0	1	0	0

Label	CRC (cont.)										Ack		EOF					
Bit #	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	
Value	0	1	1	0	1	0	0	0	1	1	0	1	1	1	1	1	1	

- b) Which device was transmitting the first frame here, and what was the data payload?

Name: _____

Problem 3 (20 points): File System

Given the initial state of a FAT filesystem and the list of accesses made starting from this state below. Assume each block is 512 bytes. Assume that blocks are grabbed from and returned back to the front of the free list on allocation and deallocation, respectively.

a) Show the state of the directory and the FAT at the end of the access sequence.

File Accesses:

1. Create File A
2. Write 700 Bytes to File A
3. Append 200 Bytes to File A
4. Append 100 Bytes to File A
5. Create File B
6. Create File C
7. Write 800 Bytes to File C
8. Write 300 Bytes to File B
9. Append 100 Bytes to File A
10. Delete File C
11. Create File D
12. Append 100 Bytes to File A
13. Write 1800 Bytes to File D
14. Append 1000 Bytes to File A

Directory:

File Name	Starting Block	File Size
X	3	2042 bytes
Y	2	256 bytes
Free	1	7,680 bytes

FAT	Initial Data	New Data
0	X	X
1	5	
2	-	
3	4	
4	8	
5	6	
6	7	
7	9	
8	10	
9	11	
10	-	
11	12	
12	13	
13	14	
14	15	
15	16	
16	17	
17	18	
18	19	
19	20	
20	-	

b) What is the internal and external fragmentation on the disk at the end of the sequence?

Name: _____

- c) In many flash devices, the number of times a particular block on disk can be written to is limited, where wear of the device causing failures is a serious problem. One way to help prevent wear is to implement wear-leveling. Explain how you could modify the current file system implementation to minimize wear?

Scratch space for Problem 3(a):

Name: _____

- b) For each of the 3 different heap algorithms and the sequence of allocations in part a), how much internal and external fragmentation has each heap allocation scheme at the end of the sequence or at the point where the sequence fails? Which heap allocation scheme had the least amount of external fragmentation (size of the largest allocatable block vs. total free space)?

- c) If any of the allocation schemes fail, is it possible to rearrange the order of mallocs and free such that all of the allocation schemes are successful? If so, provide the new order, or explain why it is not possible.

Name: _____

Problem 5 (20 points + 10 points bonus): Real-Time System Design

A group of Aggies are attempting to develop their own Formula 0001 race car with the following components:

- An external sensor that brings in data that, based on the maximum car speed, changes at a maximum rate of 125Hz.
- A sensor sampling task that reads one sample from the ADC and processes it taking 1ms every time it executes.
- A PID controller that takes 1 ms every time it executes and controls a motor with a time constant of 40ms.
- A network task that receives 18 bytes of data from the network and places them into a queue taking 2ms every time it runs, where network data arrives at a rate of 3,000 bytes/s.
- A logging task that removes 24 bytes of data from the queue and writes it to disk taking 1ms every time it runs. The SD card has a maximum bandwidth of 10 kbytes/s.

The Aggies implemented this system by running a rate monotonic scheduler with the following task periods and priorities below:

Task	Duration	Period	Priority
Logging	1 ms	12 ms	1
Control PID	1 ms	2 ms	2
ADC Sampling	1 ms	5 ms	3
Network	2 ms	8 ms	4

- a) Does this system work or did the Aggies screw up their implementation? What are the bugs with the current implementation?

Name: _____

- b) **(Extra Credit)** Modify the system into a working solution that does not have any issues and show the final system schedule below. You may not change the tasks or the external robot hardware.

